PROBLEM 29

Heavy Particle on a Tilted Cylinder

Original statement · Problems.pdf, p. 17

Unilateral Liftoff

A heavy particle of mass m is constrained to move without friction along the surface of an infinitely-long, circular cylinder of radius R. The cylinder axis (Oz) forms an angle α with the vertical direction (Oz_1). A sketch is shown on the left panel of the figure below. The position of the particle is completely specified by providing the coordinate z along the axis Oz and the angle θ as shown on the right panel. Initially, the particle is located at $z = 0, \theta = \pi/2$ and the time derivatives are $\dot{z} = v_0$ and $\dot{\theta} = \omega_0$.

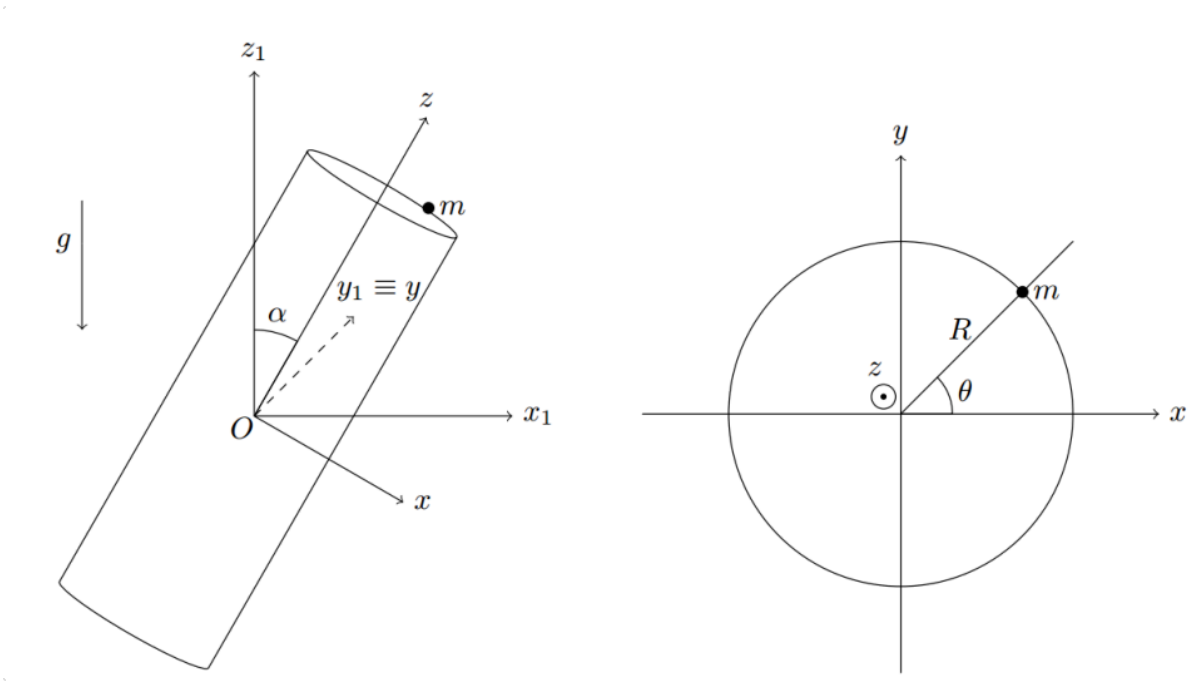


Figure: Left: cylinder with axis Oz tilted by α from the vertical Oz1; right: cross-section with radius R and angle θ (Problems.pdf, p. 17).

(a) Determine the value of z as a function of time, $z(t)$.

(b) Reduce the determination of the angle $\theta(t)$ to integrals of the form $\int[...]d\theta = \int[...]dt$.

(c) Determine the reaction from the cylinder as a function of θ.

(d) For the particular case $\omega_0^2 = \frac{g\sin\alpha}{R}$, describe qualitatively the motion of the particle.

(e) For the particular case $\omega_0^2 = \frac{4g\sin\alpha}{R}$, describe qualitatively the motion of the particle.

Worked Solution (4-Phase Methodology)

PHASE 1

Constraints, Degrees of Freedom & Coordinates

- **System:** a heavy particle of mass m moves without friction on the surface of an infinitely long circular cylinder of radius R whose axis Oz forms the angle α with the vertical Oz_1.
- **Constraint:** the particle stays on the surface, $x^2 + y^2 = R^2$ in the frame of the cylinder (holonomic, scleronomic, smooth). The statement says the particle is constrained to move along the surface, so we treat it as bilateral; the sign of the reaction tells on which side the surface presses it (checked in (c)).
- **Degrees of freedom:** $\mathrm{CDOF} = 3 - 1 = 2$, with the coordinates of the statement: the axial coordinate $z(t)$ and the angle $\theta(t)$ around the axis.
- **Initial state:** $z(0) = 0, \theta(0) = \pi/2, \dot{z}(0) = v_0, \dot{\theta}(0) = \omega_0$.
- **Parameters:** m, R, α, g, v_0, ω_0.

PHASE 2

Reference Frames & Vector Bases

- **Vertical frame:** $S_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ with $\mathbf{k}_1$ vertical upward.
- **Cylinder frame:** $S_2 = \{O; \hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}\}$ with $\hat{\mathbf{z}}$ along the axis. From the figure, $\hat{\mathbf{z}} = \sin\alpha\,\mathbf{i}_1 + \cos\alpha\,\mathbf{k}_1$, $\hat{\mathbf{x}} = \cos\alpha\,\mathbf{i}_1 - \sin\alpha\,\mathbf{k}_1$ and $\hat{\mathbf{y}} = \mathbf{j}_1$, a rotation of S_1 by α about $\mathbf{j}_1$:

$$[{}_1R_2] = \begin{pmatrix} \cos\alpha & 0 & \sin\alpha \\ 0 & 1 & 0 \\ -\sin\alpha & 0 & \cos\alpha \end{pmatrix}, \qquad \det[{}_1R_2] = \cos^2\alpha + \sin^2\alpha = 1, \qquad [{}_1R_2][{}_1R_2]^T = I$$

- **Cylindrical basis:** $\mathbf{e}_r = \cos\theta\,\hat{\mathbf{x}} + \sin\theta\,\hat{\mathbf{y}}$, $\mathbf{e}_\theta = -\sin\theta\,\hat{\mathbf{x}} + \cos\theta\,\hat{\mathbf{y}}$, $\hat{\mathbf{z}}$, a rotation of S_2 by θ about $\hat{\mathbf{z}}$ (matrix $\begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$, $\det = \cos^2\theta + \sin^2\theta = 1$; RR^T has diagonal entries $\cos^2\theta + \sin^2\theta = 1$ (and 1) and off-diagonal entries $\cos\theta\sin\theta - \sin\theta\cos\theta = 0$, so $RR^T = I$), with $\dot{\mathbf{e}}_r = \dot\theta\,\mathbf{e}_\theta$ and $\dot{\mathbf{e}}_\theta = -\dot\theta\,\mathbf{e}_r$.
- **Gravity in this basis.** From $\mathbf{k}_1 = \cos\alpha\,\hat{\mathbf{z}} - \sin\alpha\,\hat{\mathbf{x}}$:

$$-g\,\mathbf{k}_1 = g\sin\alpha\,\hat{\mathbf{x}} - g\cos\alpha\,\hat{\mathbf{z}} = g\sin\alpha\cos\theta\,\mathbf{e}_r - g\sin\alpha\sin\theta\,\mathbf{e}_\theta - g\cos\alpha\,\hat{\mathbf{z}}$$

so $\theta = 0$ is the lowest generator of the cylinder ($\mathbf{e}_r$ points downward there).

PHASE 3

Derivation

3.1 Kinematics

Pedagogical Justification: on the cylinder $r = R$ is constant, so the position uses the two free coordinates, and Poisson's formula gives the derivatives of $\mathbf{e}_r$:

$$\mathbf{r} = R\,\mathbf{e}_r + z\,\hat{\mathbf{z}}, \qquad \mathbf{v} = R\dot\theta\,\mathbf{e}_\theta + \dot z\,\hat{\mathbf{z}}, \qquad \mathbf{a} = -R\dot\theta^2\,\mathbf{e}_r + R\ddot\theta\,\mathbf{e}_\theta + \ddot z\,\hat{\mathbf{z}}$$

3.2 Forces

- **Applied force:** the weight $m\mathbf{g}$, decomposed in 2.
- **Constraint force:** the reaction of the smooth cylinder is normal to the surface, so it is radial: $\mathbf{N} = -N\,\mathbf{e}_r$ (with $N > 0$ when it pushes toward the axis). It has no component along $\hat{\mathbf{z}}$ or $\mathbf{e}_\theta$.
- **Inertia forces:** none (S_1 is inertial).

3.3 Equations of Motion

Pedagogical Justification: Newton's second law in S_1 projected on the three directions of the cylindrical basis gives three scalar equations for the three unknowns z, θ and N. The axial and azimuthal projections contain no reaction and are the equations of motion; the radial one gives N:

$$\hat{\mathbf{z}}:\ m\ddot{z} = -mg\cos\alpha, \qquad \mathbf{e}_\theta:\ mR\ddot{\theta} = -mg\sin\alpha\sin\theta, \qquad \mathbf{e}_r:\ -mR\dot{\theta}^2 = -N + mg\sin\alpha\cos\theta$$

- **Decoupling.** The axial equation involves only z and the azimuthal one only θ (a pendulum equation); the reaction is computed afterwards from the radial equation.
- **First integral.** Multiplying the azimuthal equation by $\dot{\theta}$ (chain rule, $\frac{d}{dt}\left(\frac{1}{2}\dot{\theta}^2\right) = \dot{\theta}\ddot{\theta}$ and $\frac{d}{dt}\cos\theta = -\sin\theta\,\dot{\theta}$):

$$\frac{d}{dt}\left[\frac{1}{2}\dot{\theta}^2 - \frac{g\sin\alpha}{R}\cos\theta\right] = \dot{\theta}\left(\ddot{\theta} + \frac{g\sin\alpha}{R}\sin\theta\right) = 0$$

3.4 Integration and Answers

(a) Axial motion $z(t)$

Pedagogical Justification: the axial equation is $\ddot{z} = -g\cos\alpha$. Integrating twice with Barrow's rule from the initial conditions:

$$\dot{z}(t) = v_0 + \int_0^t (-g\cos\alpha)\,dt' = v_0 - g\cos\alpha\, t, \qquad \boxed{z(t) = \int_0^t (v_0 - g\cos\alpha\, t')\,dt' = \left[v_0 t' - \tfrac{1}{2}g\cos\alpha\, t'^2\right]_0^t = v_0 t - \tfrac{1}{2}g\cos\alpha\, t^2}$$

(b) Reduction of $\theta(t)$ to quadratures

- From the first integral, with $\theta = \pi/2$ ($\cos\theta = 0$) and $\dot{\theta} = \omega_0$ at $t = 0$, the constant is $\frac{1}{2}\omega_0^2$:

$$\dot{\theta}^2 = \omega_0^2 + \frac{2g\sin\alpha}{R}\cos\theta$$

- Separating variables (Barrow's rule from $\theta(0) = \pi/2$):

$$\boxed{\int_{\pi/2}^{\theta(t)} \frac{d\theta'}{\sqrt{\omega_0^2 + \dfrac{2g\sin\alpha}{R}\cos\theta'}} = \pm t}$$

with the sign of $\dot\theta$.

(c) Reaction of the cylinder

> **Pedagogical Justification:** the radial equation gives $N = mR\dot\theta^2 + mg\sin\alpha\cos\theta$; substituting $\dot\theta^2$ from (b):

$$N(\theta) = mR\left(\omega_0^2 + \frac{2g\sin\alpha}{R}\cos\theta\right) + mg\sin\alpha\cos\theta = \boxed{mR\omega_0^2 + 3mg\sin\alpha\cos\theta}$$

N is the component toward the axis. If it becomes negative, the surface would have to pull the particle: possible for a bilateral constraint, but a particle resting on the inside of the cylinder would leave the surface.

(d) Case $\omega_0^2 = g\sin\alpha/R$

> **Pedagogical Justification:** the turning points of the angular motion are where $\dot\theta = 0$:

$$\dot\theta^2 = \frac{g\sin\alpha}{R}(1 + 2\cos\theta) = 0 \implies \cos\theta = -\frac{1}{2} \implies \theta = \pm\frac{2\pi}{3}$$

- **Motion:** since $\theta = 0$ is the stable bottom generator, the particle oscillates in θ between $-2\pi/3$ and $+2\pi/3$ about the bottom (for $\omega_0 > 0$ it first moves to $+2\pi/3$, then swings through $\theta = 0$ to $-2\pi/3$ and back), while it moves along the axis with the uniform deceleration $-g\cos\alpha$: $z(t) = v_0 t - \frac{1}{2}g\cos\alpha\, t^2$. The trajectory is a wavy curve on the cylinder.
- **Reaction:** $N = mg\sin\alpha\,(1 + 3\cos\theta)$, which is negative for $\cos\theta < -1/3$ ($\theta > 109.5°$), so the motion up to $\pm 2\pi/3$ (where $N = -\frac{1}{2}mg\sin\alpha$) needs a bilateral constraint.

(e) Case $\omega_0^2 = 4g\sin\alpha/R$

> **Pedagogical Justification:** now $\dot\theta^2 = \dfrac{g\sin\alpha}{R}(4+2\cos\theta) \geq \dfrac{2g\sin\alpha}{R} > 0$ for every θ (the minimum is at $\theta = \pi$, the top).

- **Motion:** the angular velocity never vanishes and keeps its sign, so the particle makes complete turns around the cylinder, in a helical-like trajectory whose pitch follows the axial velocity $\dot z = v_0 - g\cos\alpha\, t$, which decreases, vanishes at $t = v_0/(g\cos\alpha)$ and then reverses sign, so the pitch shrinks to zero and then changes sense. The reaction $N = mg\sin\alpha\,(4+3\cos\theta) \geq mg\sin\alpha > 0$ is always positive, so this motion is also admissible if the particle is inside a hollow cylinder.

PHASE 4

Interpretation & Checks

- **Decoupling:** the axial motion is a uniformly decelerated one-dimensional motion and the azimuthal motion is that of a nonlinear pendulum of length R in the effective gravity $g\sin\alpha$ (the component of gravity in the cross-section).
- **Dimensional check:** $[g\sin\alpha/R] = \mathrm{s}^{-2} = [\omega_0^2]$ and $[N] = [mR\omega_0^2] = \mathrm{N}$.
- **Limits:** for $\alpha \to 0$ (vertical axis) there is no azimuthal force and $\dot\theta = \omega_0$; for $\alpha \to \pi/2$ (horizontal axis) $\ddot z \to 0$.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 29 in its solution keys, so the results are verified by independent derivation. Sympy reproduces the components of the acceleration in the cylindrical basis and those of gravity ($\mathbf{g}\cdot\hat{\mathbf{z}} = -g\cos\alpha$, $\mathbf{g}\cdot\mathbf{e}_r = g\sin\alpha\cos\theta$, $\mathbf{g}\cdot\mathbf{e}_\theta = -g\sin\alpha\sin\theta$). Numerical integration of $\ddot\theta = -(g\sin\alpha/R)\sin\theta$ from $\theta(0) = \pi/2$ gives the turning points $\pm 0.66667\pi = \pm 2\pi/3$ for (d), no turning point for (e), and a minimum of N of $-0.5\,mg\sin\alpha$ for (d) and $+1.0\,mg\sin\alpha$ for (e).

Source: Problems.pdf